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Cloud Computing

Elective

5th Semester CSE

UNIT-I

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Part 1: Introduction to Cloud Computing

Text Book:

Cloud Computing: A Comprehensive Guide

Defining Types, Characteristics, Standards, and Services

Based on Barrie Sosinsky's "Cloud Computing Bible"

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Agenda & Learning Objectives

- Understand the core definition and evolution of Cloud Computing.
- Identify and differentiate the four primary Cloud Deployment Models.
- Analyze the five essential Characteristics of Cloud Computing.
- Assess the critical Role of Open Standards.
- Define, compare, and contrast the core Service Models: IaaS, PaaS, SaaS.
- Explore specialized services: IDaaS and CaaS.
- Discuss the Business Impact and Future Directions.

What is Cloud Computing? - A Definition

- **Sosinsky's Definition:** "Cloud computing is a system of pooled computing resources that are delivered over a network, most often the Internet."
- **NIST Definition:** "A model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources..."
- **Core Idea:** Moving from owning infrastructure to renting services.

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The Historical Context & Evolution

- Mainframe and Time-Sharing (1960s)
- The PC Revolution (1980s)
- The Rise of the Internet and Dot-com Era (1990s)
- Grid Computing and Utility Computing Concepts
- The Perfect Storm: Broadband, Web 2.0, Virtualization (2000s)
- The Advent of Modern Cloud Providers (AWS, Google, Azure)

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Why Cloud? The Business Driver

- **Shift from CapEx to OpEx:** No large upfront investment in hardware; pay-as-you-go.
- **Speed and Agility:** Drastically reduced time to market for new applications.
- **Focus on Core Business:** Instead of managing complex IT infrastructure.

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Cloud Deployment Models (Cloud Types)

Understanding Cloud Deployment Models

- Defined by *who* has access to the cloud infrastructure.
- Four primary models as defined by NIST and covered by Sosinsky.

Public Cloud

- **Definition:** Services offered over the public Internet to the general public or a large industry group.
- **Ownership:** Owned and operated by a third-party cloud service provider.
- **Key Traits:** Massive scale, cost-effective, high reliability.
- **Examples:** Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP).
- **Use Case:** Web hosting, development environments, SaaS applications.

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Public Cloud - Advantages & Disadvantages

- **Advantages:**
  - Lower Costs (No CapEx)
  - No Maintenance
  - High Scalability & Elasticity
  - High Reliability
- **Disadvantages:**
  - Potential Security & Compliance Concerns
  - Less Control over Infrastructure
  - Potential for Vendor Lock-in

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Private Cloud

- **Definition:** Cloud infrastructure operated solely for a single organization.
- **Ownership:** Can be managed internally or by a third party, on or off-premise.
- **Key Traits:** Enhanced security, control, and customization.
- **Examples:** VMware vSphere, OpenStack, Microsoft Hyper-V.
- **Use Case:** Banks, government agencies, companies with strict data security needs.

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Private Cloud - Advantages & Disadvantages

- **Advantages:**
  - Greater Control & Customization
  - Enhanced Security & Compliance
  - Predictable Performance
- **Disadvantages:**
  - Higher CapEx and IT Operational Costs
  - Limited Scalability compared to Public cloud
  - Requires In-House IT Expertise

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Community Cloud

- **Definition:** Infrastructure shared by several organizations from a specific community with common concerns.
- **Ownership:** Could be managed by the organizations or a third party.
- **Key Traits:** Shared costs, shared compliance requirements.
- **Examples:** A cloud for all banks in a country, a cloud for government agencies, a research cloud for universities.
- **Use Case:** Industries with shared regulatory frameworks

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Hybrid Cloud

- **Definition:** A composition of two or more distinct cloud infrastructures (public, private, community) that remain unique entities but are bound together by standardized technology.
- **Key Technology:** Cloud Bursting, VPNs, APIs.
- **Goal:** To create a unified, flexible environment.
- **Use Case:** Running a core application on a private cloud but "bursting" to the public cloud for peak loads.

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Hybrid Cloud - Use Case Diagram

- **On-Premise (Private Cloud):** Sensitive customer data, core transactional database.
- **Hybrid Connection:** Secure, encrypted network link.
- **Public Cloud:** Web servers, development/test environments, data analytics during sales peaks.
- **Caption:** "Bursting" capacity to the public cloud as needed.

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Comparing the Four Cloud Types

A quick-reference table

Model	Access	Control	Cost	Scalability	Best For
Public	Everyone	Low	OpEx	Very High	Startups, Web Apps
Private	Single Org	High	CapEx	Limited	Regulated Industries
Community	Member Orgs	Medium	Shared	Medium	Specific Industries
Hybrid	Mixed	Mixed	Mixed	High	Dynamic Workloads

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Core Characteristics of Cloud Computing

The Five Essential Characteristics (NIST)

According to NIST, a cloud service must exhibit these five traits.

On-Demand Self-Service

- **Definition:** A consumer can unilaterally provision computing capabilities (e.g., server time, network storage) automatically without requiring human interaction with the service provider.
- **Example:** Clicking a button in the AWS console to launch a new virtual server in minutes.

Broad Network Access

- **Definition:** Capabilities are available over the network and accessed through standard mechanisms (e.g., mobile phones, tablets, laptops, workstations).
- **Principle:** The "Network is the Computer." Access from anywhere

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Resource Pooling

- **Definition:** The provider’s computing resources are pooled to serve multiple consumers using a multi-tenant model.
- **What's Pooled:** Physical and virtual resources are dynamically assigned and reassigned according to consumer demand.
- **Location Independence:** The customer generally has no control or knowledge over the exact location of the provided resources.

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Rapid Elasticity

- **Definition:** Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand.
- **To the Consumer:** Capabilities often appear to be unlimited and can be appropriated in any quantity at any time.
- **Example:** An e-commerce website scaling up for Black Friday and scaling down afterward automatically.

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Measured Service

- **Definition:** Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service.
- **Pay-per-Use:** Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer.
- **Examples:** Paying for compute per hour, storage per GB, bandwidth per GB.

Characteristics Summary

- These five characteristics are what differentiate true cloud computing from traditional hosting or outsourcing.
- They enable the economic and technical benefits of the cloud model.

The Role of Open Standards

The Importance of Open Standards

- **The Problem:** Without standards, each cloud provider creates a unique, proprietary environment.
- **The Goal:** Interoperability, Portability, and Choice.

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What are Open Standards?

- **Definition:** Publicly available specifications for achieving interoperability and data exchange.
- They are developed collaboratively and are not controlled by a single vendor.
- **Key Areas:**
  - **APIs:** How applications talk to the cloud.
  - **Data Formats:** How data is stored and exchanged (XML, JSON).
  - **Protocols:** How data is transported (HTTP, TCP/IP).

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The Risk of Vendor Lock-In

- **Definition:** High dependence on a single cloud provider's specific tools, APIs, and services.
- **Consequence:** It becomes technically difficult and expensive to move to another provider.
- **Sosinsky's Warning:** This can create significant long-term costs and strategic limitations.

Strategies for Using Open Standards

- Prefer services that support standard APIs (e.g., S3-compatible storage).
- Use open data formats instead of proprietary ones.
- Leverage containerization (e.g., Docker, Kubernetes) for application portability.
- Implement cloud-agnostic management tools.

Core Service Models (The Cloud Stack)

The "As-a-Service" Spectrum

- Cloud services are layered, forming a stack of abstraction.
- The three core models define the division of responsibility between the user and the provider.

The Shared Responsibility Model

- A fundamental concept in cloud computing.
- **The Provider** is always responsible for **security *of* the cloud** (the underlying infrastructure).
- **The User** is always responsible for **security *in* the cloud** (their data, access, applications).
- The division of everything else depends on the service model.

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Visual: The Cloud Stack Analogy

- **On-Premise:** You manage everything. (The whole pizza)
- **IaaS:** The provider gives you the kitchen and oven. You bring the ingredients and cook. (Pizza base & toppings)
- **PaaS:** The provider gives you a pre-heated oven and pre-made dough. You just add toppings. (Just the toppings)
- **SaaS:** The provider gives you a fully cooked pizza. You just eat it. (The finished product)

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Defining Infrastructure as a Service (IaaS)

- **What it is:** The most fundamental building block. Provides raw computing infrastructure over the Internet.
- **Core Components:**
  - Virtual Machines (Compute)
  - Storage (Block, Object)
  - Networks (Virtual Networks, Firewalls)
  - Operating Systems (User's choice)

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IaaS - Division of Responsibility

- **Provider Manages:** Virtualization, Servers, Storage, Networking.
- **User Manages:** Operating System, Runtime, Middleware, Applications, and Data.
- **Analogy:** Renting a data center in someone else's building.

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IaaS - Key Use Cases

- **Web Hosting:** Full control over the web server environment.
- **Development & Testing:** Quickly spin up and tear down environments.
- **Big Data Analysis:** Massive, scalable compute clusters for data processing.
- **Disaster Recovery:** Replicating on-premise servers to the cloud.

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IaaS - Major Providers & Examples

- **Amazon EC2 & S3** (Elastic Compute Cloud & Simple Storage Service)
- **Google Compute Engine & Cloud Storage**
- **Microsoft Azure Virtual Machines & Blob Storage**
- **Rackspace**

Defining Platform as a Service (PaaS)

- **What it is:** Provides a platform and environment for developers to build, test, deploy, and manage applications without the complexity of building and maintaining the underlying infrastructure.
- **Goal:** Streamline the developer experience.

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PaaS - Division of Responsibility

- **Provider Manages:** Runtime, Middleware, Operating System, Virtualization, Servers, Storage, Networking.
- **User Manages:** Applications and Data.
- **Analogy:** Renting a fully-equipped workshop.

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PaaS - Key Use Cases

- **Application Development & Deployment:** Streamlined DevOps lifecycle.
- **API Development and Management:** Building and scaling APIs.
- **Internet of Things (IoT):** Platforms for handling IoT data streams.
- **Business Intelligence:** Tools for data analytics and visualization.

PaaS - Major Providers & Examples

- Google App Engine
- Microsoft Azure App Service
- Salesforce Heroku
- AWS Elastic Beanstalk

Defining Software as a Service (SaaS)

- **What it is:** Software that is licensed on a subscription basis and is centrally hosted. Accessed via a web browser.
- **Sometimes called:** "On-Demand Software" or "Web-Based Software."

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SaaS - Division of Responsibility

- **Provider Manages:** Everything: Applications, Data, Runtime, Middleware, O/S, Virtualization, Servers, Storage, Networking.
- **User Manages:** User-specific configuration and their own data.
- **Analogy:** Using a taxi service vs. owning a car.

SaaS - Key Use Cases

- **Enterprise Applications:** CRM, ERP, Email, Collaboration.
- **Personal Productivity:** Office Suites, File Storage.
- **Vertical Market Software:** Applications for specific industries (e.g., medical records).

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SaaS - Major Providers & Examples

- **Google Workspace** (Gmail, Docs, Drive)
- **Microsoft 365** (Outlook, Word, Teams)
- **Salesforce** (CRM)
- **Dropbox, Slack, Zoom**

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Comparing IaaS, PaaS, SaaS

Model	User Control	User Responsibility	Provider Responsibility	Analogy
IaaS	High	OS, Apps, Data	Virtualization & Below	Rental Kitchen
PaaS	Medium	Apps, Data	Runtime & Below	Rented Workshop
SaaS	Low	Data & Config	Everything	Taxi Service

Specialized Cloud Services

Beyond the Core: The "XaaS" Universe

- Many other capabilities are now delivered "as-a-Service."
- Two critical ones for security and compliance are IDaaS and CaaS.

Defining Identity as a Service (IDaaS)

- **What it is:** Cloud-based authentication and identity management provided as a service.
- **Core Problem:** Managing user identities and access across a growing number of cloud and on-premise applications is complex.

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IDaaS - Core Functions & Benefits

- **Functions:**
  - Single Sign-On (SSO)
  - Multi-Factor Authentication (MFA)
  - User Lifecycle Management (Provisioning/De-provisioning)
  - Directory Services
- **Benefits:**
  - Improved Security & Compliance
  - Enhanced User Experience (One password)
  - Reduced IT Overhead

IDaaS - Major Providers & Examples

- **Okta**
- **Ping Identity**
- **Microsoft Azure Active Directory**
- **OneLogin**

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Defining Compliance as a Service (CaaS)

- **What it is:** A service where a cloud provider ensures its infrastructure meets specific regulatory requirements and offers tools to help customers demonstrate their own compliance.
- **Core Problem:** Audits and compliance certifications are expensive and time-consuming.

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CaaS - Common Compliance Frameworks

- **PCI-DSS:** For organizations handling credit card data.
- **HIPAA:** For protecting sensitive patient health information.
- **SOC 1/2/3:** Reports on controls at a service organization.
- **GDPR:** For data protection and privacy in the European Union.
- **FedRAMP:** For U.S. federal government data.

CaaS - How it Works & Benefits

- **How it Works:**

  The provider certifies its data centers and core services.

  The provider offers compliance-oriented tools (e.g., logging, monitoring, policy templates).

  The customer uses these certified building blocks to build their own compliant systems.

- **Benefit:** Drastically reduces the compliance burden for the customer.

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Conclusion & The Future

The Business Impact: Summary of Benefits

- **Cost Efficiency:** OpEx model, no hardware maintenance.
- **Scalability & Flexibility:** Instantly scale to meet demand.
- **Strategic Focus:** Focus on your business, not your data center.
- **Reliability & Disaster Recovery:** Provider-level SLAs and global infrastructure

Challenges & Considerations

- **Security & Compliance:** Shared responsibility model must be understood.
- **Vendor Lock-In:** A persistent risk that must be managed.
- **Network Dependency:** Requires a reliable, high-speed Internet connection.
- **Potential for Uncontrolled Costs:** Requires careful monitoring and governance

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The Future of Cloud Computing

- **Serverless Computing:** Abstracting away servers entirely (e.g., AWS Lambda).
- **Edge Computing:** Processing data closer to where it's created.
- **AI/ML as a Service:** Democratizing access to artificial intelligence.
- **Quantum Computing as a Service:** Early access to quantum power via the cloud.
- **Ubiquitous Hybrid/Multi-Cloud:** The norm for enterprise IT.

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Key Takeaways

- Cloud is a transformative model for delivering IT.
- Choose the right *Deployment Model* (Public, Private, Hybrid) for your needs.
- Understand the *Service Models* (IaaS, PaaS, SaaS) and the shared responsibility.
- Prioritize *Open Standards* to maintain flexibility.
- Leverage specialized services like *IDaaS* and *CaaS* to solve complex problems.